PV102 Day 1: Applications and Design Decisions

Course transition from verified workmanship to system-level judgment

CTS Day 17	10:00 AM-3:00 PM	6 students

Instructor Run Sheet | Audience: Instructor or substitute | Monday, August 3, 2026

Core rule

Start with load, site, and need. PV101 proved that students can assemble and verify a small system. PV102 begins by deciding which system belongs in the job.

Learning targets

- Distinguish grid-tied, hybrid, off-grid, and PV-direct architectures.
- Trace a first-pass energy path from PV source to the intended load.
- Identify load, site, access, budget, maintenance, and resilience constraints.
- Defend an architecture choice without guessing beyond the available evidence.
- State protection, isolation, and verification boundaries before construction.

Readiness and materials

Before students arrive	Materials
• Open slide deck and test display. • Print six quizzes, one student packet per student, and one lab scenario per team. • Set three team stations with chart paper, markers, calculators, and component cards. • Cover or isolate PV modules; keep batteries and electrical hardware disconnected.	• Projector and slide deck • Printed quizzes and packets • Three component-card sets or de-energized components • Chart paper, markers, calculators, clipboards • PPE only if students handle physical components

Five-hour run of show

Time	Block	Instructor move
10:00-10:15	**PV102 transition**	Name the progression: verified workmanship now supports design judgment. Introduce the core rule.
10:15-10:40	**Morning quiz and review**	Students work independently for 12-15 minutes. Review by asking for evidence, not just letters.
10:40-11:30	**Applications lesson**	Use slides 1-16. Pause after each architecture and ask what client need it serves.
11:30-12:15	**Guided architecture sort**	Read three client prompts aloud. Students choose an architecture, then revise when a new constraint is revealed.
12:15-12:45	**Lunch**	Reset three lab stations. Keep all electrical equipment disconnected.
12:45-1:00	**Lab brief**	Assign teams and roles. Model the acceptance criteria and the two-minute handoff.
1:00-2:15	**Scenario design lab**	Teams complete one primary scenario. Rotate if finished; instructor probes assumptions and safety boundaries.
2:15-2:40	**Handoffs and peer verification**	Each team gives a two-minute handoff. Peers check need, path, constraint, boundary, and unknowns.
2:40-3:00	**Exit ticket and reset**	Collect packets. Preview Day 2: site and solar-resource assessment.

Talk track: course transition

Opening script

"You already know how to connect, inspect, measure, and hand off a small PV system. PV102 asks a harder question: what system should exist in the first place? A professional starts with the client's load, the site, and the real operating need."

Architecture teaching notes

Architecture	Ask students	Listen for
Grid-tied	What job does the utility perform?	Normal grid support; bill reduction; no assumed outage operation
Hybrid	Which exact loads need backup, and for how long?	Defined resilience goal; storage has a specific job
Off-grid	What happens through the night or poor weather?	Load continuity, autonomy, service access, and storage
PV-direct	Can the load wait for sun?	Flexible timing or process storage such as a water tank

Guided architecture sort

1. Start with: "The client wants solar." Reject this as incomplete. Ask: what must the system accomplish?
2. Reveal a utility connection. Students choose between grid-tied and hybrid only after the outage requirement is stated.
3. Reveal a remote site. Ask whether the load must operate at night; use that answer to distinguish off-grid from PV-direct.
4. For every answer, require one sentence of evidence and one unknown that still needs verification.

Lab coaching prompts

- What must run, and when must it run?
- What does the site allow or prevent?
- What job would a battery perform here?
- Where must energy be isolated or overcurrent limited?
- Which statement is verified, and which is still an assumption?
- What would make you change this architecture choice?

Common misconceptions and corrections

Misconception	Correction
Every solar system needs batteries.	Storage is selected only when it performs a defined job.
Grid-tied solar automatically powers a building during an outage.	Outage operation requires equipment and design specifically intended for that function.
The biggest array is the best design.	The best design matches the load, site, goals, and constraints.
Protection can be added after the layout is complete.	Protection, isolation, labeling, and access belong in the first design pass.
A missing fact can be replaced with a reasonable guess.	Record the unknown and identify how it will be verified.

Assessment evidence

- Quiz: prerequisite electrical and verification habits.
- Design brief: architecture choice, energy path, constraints, boundaries, and unknowns.
- Handoff: scope, decision, evidence, safety boundary, and unresolved items.
- Exit ticket: operating need, storage decision, and Day 2 verification question.

Minimum successful performance

The student selects a defensible architecture, maps a plausible major-component path, cites scenario evidence, names a protection or isolation boundary, and records unknowns instead of guessing.

Answer snapshot and contingency plan

Scenario anchors

- Remote trail counter: off-grid; day-and-night operation and no utility support make storage and serviceability central.
- Small shop: hybrid; grid-connected bill reduction plus a defined limited-backup job.
- Daytime water pump: PV-direct is the strongest first-pass choice; the tank provides process storage and night operation is unnecessary.

If equipment or printing is unavailable

- Run the entire lab with handwritten component cards. No electrical hardware is required.
- Project the scenarios and have teams build one design brief on chart paper.
- If the slide deck cannot open, use the Whiteboard Outline and the architecture reference table.
- If time is short, complete one scenario deeply and use the other two as oral checks.

Early-finisher extension

- Add one new client constraint that forces a redesign.
- Compare a no-storage option with a storage option and name the job the battery performs.
- Create a verification plan for every unknown in the brief.

Safety and scope boundaries

No energized work

This lesson does not authorize live electrical work, utility interaction, battery connection, PV module connection, or code-compliance approval. Use covered/isolated modules and disconnected components only. Follow manufacturer instructions and the program's established electrical safety procedures in later hands-on work.

Day 2 handoff

Carry forward the three design briefs and their unknowns. Day 2 will convert vague site assumptions into observations: solar access, shade, orientation, space, weather, access, and maintenance conditions.

Instructional sources

- U.S. Department of Energy, Solar Energy: https://www.energy.gov/topics/solar-energy
- U.S. Energy Information Administration, Solar explained: https://www.eia.gov/energyexplained/solar/
- OSHA, Green Job Hazards - Solar Energy: Electrical: https://www.osha.gov/green-jobs/solar/electrical
- OSHA, Green Job Hazards - Solar Energy: Lockout/Tagout: https://www.osha.gov/green-jobs/solar/lockout-tagout
- Sean White, Solar Photovoltaic Basics, 2nd ed., Routledge, 2018, ISBN 9781138102866.

PV102 Day 1 Morning Quiz: Answer Key

Scoring notes and instructional rationales

10 points	Suggested: 8 ready	Use for review

Morning Quiz - Answer Key | Audience: Instructor | Monday, August 3, 2026

Item	Answer	Rationale / look-for
1	B. DC	PV modules produce direct-current electrical output.
2	A. Voltage	In a series connection, matching module voltages add while current is limited by the string.
3	B. Current	In a parallel connection, matching string currents add while voltage remains at the branch voltage.
4	C. Battery to controller	For the training hardware used in PV101, students followed the controller manufacturer's battery-first startup sequence. Always use the actual manufacturer's procedure.
5	B. Limit damaging current	A fuse or breaker opens the circuit when current exceeds its intended limit; it does not create or store energy.
6	B. De-energized and verified	Continuity and resistance checks require an electrically safe, de-energized circuit.
7	24 W	P = V x I = 12 V x 2 A = 24 W.
8	Any two: scope/work completed; verified measurements or results; condition/status; unresolved items; safety boundaries; next action.	A professional handoff separates what is complete and verified from what remains open.
9	Stop work, protect the area, verify the condition with the correct procedure/tool, and ask the instructor or qualified person.	Uncertainty is a stop condition, not permission to guess.
10	Accept any accurate statement: component ratings, functions, environment, compatibility, or client goals may not match the specific system.	PV102 shifts from identifying parts to matching a whole architecture to a real job.

Use the review diagnostically

- If items 2-3 are weak, review series and parallel behavior before Day 3 module analysis.
- If items 4-6 or 9 are weak, reinforce de-energization, manufacturer sequence, protection, and stop-work expectations before any later hands-on activity.
- If item 8 is weak, use the RCS handoff pattern: scope, verified evidence, unresolved items, next action.
- Do not treat this quiz as evidence that a student is qualified for live electrical work.

PV102 Day 1 Whiteboard Outline

Use when presenting without the slide deck or to anchor discussion

Board 1: job	Board 2: path	Board 3: proof

Whiteboard Outline | Audience: Instructor | Monday, August 3, 2026

Board 1 - Start with the job

- LOAD: what must run?
- TIME: when and for how long?
- SITE: what helps or blocks the system?
- NEED: bill reduction, resilience, remote service, or process operation?

Board 2 - Four architectures

- Grid-tied: PV + utility
- Hybrid: PV + storage + utility
- Off-grid: PV + storage + load
- PV-direct: PV + compatible daytime load

Board 3 - Design acceptance

- Match the need.
- Show the energy path.
- Name the dominant constraint.
- State the safety boundary.
- Record unknowns; do not guess.

Scenario reveals

Scenario	First-pass architecture	Decision signal
Remote trail counter	Off-grid	No utility; must run day and night
Small shop	Hybrid	Bill reduction plus defined backup loads
Daytime pump	PV-direct	Load can wait for sun; water tank stores the service

Final board sentence

The best PV system is the one matched to its real job.

PV102 Day 1 Student Packet

Applications and design decisions

Quiz | Reference | Design lab

Student Print Packet | Audience: Student | Monday, August 3, 2026

Today's rule

Start with load, site, and need. Record unknowns instead of guessing.

Learning targets

- Distinguish four common PV architectures.
- Map a first-pass energy path.
- Use client and site evidence to defend a design choice.
- State protection and verification boundaries before construction.

Morning quiz

1. What type of electrical output is produced directly by a PV module?

A. AC
B. DC
C. Mechanical
D. Thermal

2. When matching PV modules are connected in series, which quantity primarily adds?

A. Voltage
B. Current
C. Sunlight
D. Frequency

3. When matching PV strings are connected in parallel, which quantity primarily adds?

A. Voltage
B. Current
C. Resistance only
D. Frequency

4. For the small battery-based training systems used in PV101, which connection should be made first when the controller manufacturer's procedure requires battery-first startup?

A. PV to controller
B. Load to inverter
C. Battery to controller
D. Ground rod to roof

5. What is the main job of an overcurrent protective device such as a fuse?

A. Increase voltage
B. Limit damaging current
C. Create sunlight
D. Store energy

6. Before using continuity mode, the circuit must be:

A. Energized
B. De-energized and verified
C. Connected to the grid
D. Under full load

7. A 12 V load draws 2 A. What is its power in watts?

_____

8. Name two items that belong in a professional technical handoff.

_____

_____

9. A teammate is unsure whether a circuit is safe to touch. What is the correct next action?

_____

_____

10. In one sentence, explain why a correct component can still be wrong for a specific system.

_____

_____

Application architecture reference

Architecture	Utility role	Storage role	Typical decision signal
Grid-tied	Available and interconnected	Often none	Bill reduction is central; outage operation is not required
Hybrid	Available	Selected backup or operating goals	Grid benefits plus defined resilience needs
Off-grid	Unavailable	Carries loads outside solar hours	Remote loads must operate without utility support
PV-direct	Unavailable or irrelevant	Usually none	Load can operate only when solar power is available

Design order

1. Define the job: what must the system accomplish?
2. Define the load: what must run, at what power, and for how long?
3. Read the site: solar access, shade, weather, space, access, and maintenance.
4. Choose a first-pass architecture.
5. Map the energy path and name required protection and isolation points.
6. Record unknowns; verify them before sizing or building.

Safety boundary

Today's work is design-only. Use paper component cards or de-energized components. Do not connect modules, batteries, controllers, inverters, or utility circuits.

Afternoon design lab

Mission

Choose the PV architecture that best matches each client's real job. Show the energy path, identify the dominant constraint, state the safety boundary, and deliver a two-minute handoff.

Team structure

- Work in three teams of two. Start with the assigned scenario, then rotate if time permits.
- Roles: designer and verifier/recorder. Switch roles after each scenario.
- Use component cards or de-energized components only. No electrical connections are permitted.

Required deliverables

1. Architecture choice: grid-tied, hybrid, off-grid, or PV-direct.
2. A labeled energy-path diagram from source to load.
3. The critical load or operating need.
4. Three site or client constraints and the dominant constraint.
5. Protection, isolation, or labeling boundaries that must be designed before construction.
6. At least two unknowns that require verification.
7. A two-minute handoff: scope, decision, evidence, safety boundary, and unresolved items.

Acceptance criteria

Match the need. Show the path. Name the constraint. State the safety boundary. Record the unknowns.

Scenario 1 - Remote trail counter

- A 12 V electronic trail counter uses 8 W continuously.
- No utility service is available.
- The site has seasonal shade and can go five days between service visits.
- The counter must collect data day and night.

Design brief

1. Architecture choice and one-sentence reason

2. Critical load or operating need

3. Energy path - write components in order and draw arrows

4. Dominant constraint and supporting evidence

5. Two additional constraints

6. Required protection, isolation, and labeling boundaries

7. Unknowns to verify before sizing or building

8. Two-minute handoff notes

Scenario 2 - Small shop

- Utility service is available.
- The owner wants lower electric bills.
- During an outage, internet equipment and office lighting totaling about 400 W should operate for four hours.
- Roof area is available but one section receives afternoon shade.

Design brief

1. Architecture choice and one-sentence reason

2. Critical load or operating need

3. Energy path - write components in order and draw arrows

4. Dominant constraint and supporting evidence

5. Two additional constraints

6. Required protection, isolation, and labeling boundaries

7. Unknowns to verify before sizing or building

8. Two-minute handoff notes

Scenario 3 - Daytime water pump

- A 300 W DC pump fills an elevated tank.
- No utility service is available.
- The pump only needs to run when strong sunlight is available.
- The tank provides water after sunset; battery storage is not requested.

Design brief

1. Architecture choice and one-sentence reason

2. Critical load or operating need

3. Energy path - write components in order and draw arrows

4. Dominant constraint and supporting evidence

5. Two additional constraints

6. Required protection, isolation, and labeling boundaries

7. Unknowns to verify before sizing or building

8. Two-minute handoff notes

Peer verification and exit ticket

Peer verification checklist

☐ The architecture matches the client's stated operating need.

☐ The energy path includes every major required function.

☐ The dominant constraint is supported by scenario evidence.

☐ Protection and isolation are treated as design requirements.

☐ Unknowns are recorded instead of guessed.

☐ The handoff separates verified facts from open questions.

Exit ticket

1. Which operating need most strongly determines architecture?

2. When is storage unnecessary?

3. What is one question you must verify before Day 2 site assessment?